1. I2S0/TDM PROGRAM-AUDIO LVDS LINK

J901 mates with AUDIO-8X8 J101. Route five 100 ohm differential pairs with the assigned interleaved returns.

SN65LVDT2D includes the ADC-data receiver termination; do not add a duplicate far-end resistor.

2. I2S1 HEADSET LEVEL TRANSLATION

PCA9306 VREF2 and EN share HS_I2C_BIAS through the required 200 k pull-up to 3.3 V.

CM5 audio GPIO is 3.3 V. ES8316 digital pins remain entirely in the translated 1.8 V domain.

3. DEDICATED LOW-NOISE HEADSET RAILS

R900 is the only headset analog-ground bond. Do not create a second PCB or chassis bond.

4. ES8316 I2S1 HEADSET CODEC

CE is pulled to 1.8 V; all ES8316 digital pins are translated. Decoupling follows the Radxa reference topology.

5. TPA6132A2 HEADPHONE AMPLIFIER

G0=1 and G1=0 select 0 dB gain. EN remains software-controlled for pop-free mute.

6. CTIA HEADSET JACK, MIC BIAS, DETECT, AND ESD

CTIA contact map: tip=L, ring1=R, ring2=GND, sleeve=MIC. Verify isolated switch polarity on a Kycon sample.

J910 is routable for prototype capture, but its drawing-derived footprint is blocked from production until sample/coupon sign-off.

7. HEADSET CONTROL POLICY

HEADSET_POWER_EN and HEADSET_AMP_EN default low so the codec and amplifier remain muted during boot.

Sequence: enable rails, configure ES8316, enable TPA6132A2, then ramp software volume.

Firmware must debounce HS_JACK_DET_N and perform pop-free mute before any route change.

Kycon STX-353K7A is CTIA-mapped; its preliminary land pattern requires a coupon
TPA6132A2 provides 0 dB dedicated stereo headphone amplification
I2S1 uses explicit 3.3 V to 1.8 V translation before the ES8316 codec
I2S0/TDM program audio is buffered over LVDS to the separate AUDIO-8X8 board
ProComm
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